

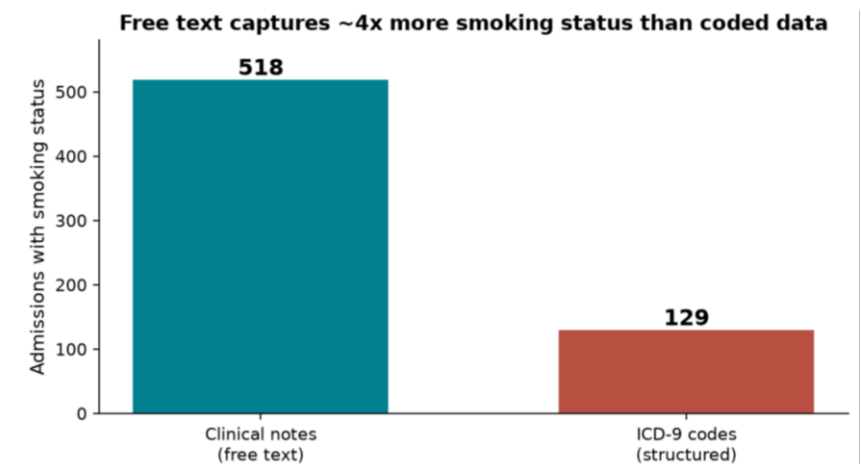
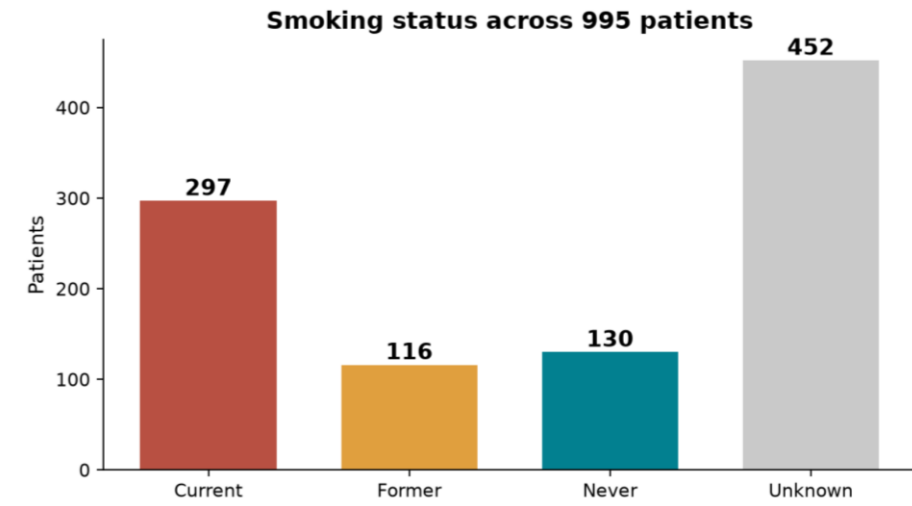
Extracting Patient Smoking Status From Clinical Text

Michael Todd



The Smoking Signal Is in the Text, Not the Codes

- **Targeted the high-yield data** - processed 1,069 discharge summaries (where social/smoking history lives) rather than all 34,619 notes
- **Ran MedCAT** with the smoking_findings_only model pack to extract tobacco/smoking SNOMED concepts
- **Cleaned & classified** mentions into current / former / never - filtering out negations, discharge advice, and false positives
- **Joined structured ICD-9 codes** as an independent second source, then resolved to one status per patient



Running MedCATtrainer

MedCAT
Projects
Metrics
Concepts
Try Model
v2.23.1
(admin)
logout

Example Project - SNOMED CT All

0-81322 | 20 Remaining

Clinical Notes

EXAM-MRI LEFT KNEE WITHOUT CONTRAST,CLINICAL;This is a 53-year-old female

REASON FOR CONSULTATION: Left hip fracture,HISTORY OF PRESENT ILLNESS: The

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HISTORY OF PRESENT ILLNESS: The patient is a 57-year-old female being seen today for evaluation of

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DIAGNOSIS: Synovitis/anterior cruciate ligament tear of the left knee,HISTORY: The

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EXAM-MRI LEFT KNEE WITHOUT CONTRAST,CLINICAL;This is a 53-year-old female with left knee pain being evaluated for ACL tear.FINDINGS;This examination was performed on 10-14-05,Normal medial meniscus without intrasubstance degeneration, surface fraying or discrete meniscal tear,There is a discoid lateral meniscus and although there may be minimal superficial fraying along the inner edge of the body, there is no discrete tear (series #6 images #7-12),There is a near-complete or complete tear of the femoral attachment of the anterior cruciate ligament. The ligament has a balled-up appearance consistent with at least partial retraction of most of the fibers of the ligament. There may be a few fibers still intact (series #4 images #12-14; series #5 images #12-14). The tibial fibers are normal,Normal posterior cruciate ligament,There is a sprain of the medial collateral ligament, with mild separation of the deep and superficial fibers at the femoral attachment (series #7 images #6-12). There is no complete tear or discontinuity and there is no meniscocapsular separation,There is a sprain of the lateral ligament complex without focal tear or discontinuity of any of the intraarticular components,Normal iliotibial band,Normal quadriceps and patellar tendons,There is contusion within the posterolateral corner of the tibia. There is also contusion within the patella at the midline patellar ridge where there is an area of focal chondral flattening (series #8 images #10-13). The medial and lateral patellar facets are otherwise normal as is the femoral trochlea in the there is no patellar subluxation,There is a mild strain of the vastus medialis oblique muscle extending into the medial patellofemoral ligament and medial patellar retinaculum but there is no complete tear or discontinuity,Normal lateral patellar retinaculum. There is a joint effusion and plica,IMPRESSION; Discoid lateral meniscus without a tear although there may be minimal superficial fraying along the inner edge of the body. Near-complete if not complete tear of the femoral attachment of the anterior cruciate ligament. Medial capsule sprain with associated strain of the vastus medialis oblique muscle. There is focal contusion within the patella at the midline patella ridge. Joint effusion and plica.

Correct

Incorrect

Terminate

Alternative

Submit

Concept Summary

Annotated Text

n/a

Name

Type IDs

n/a

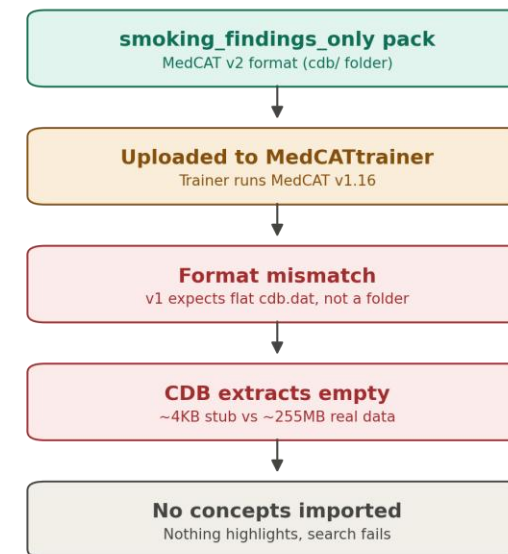
Concept ID

n/a

Accuracy

n/a

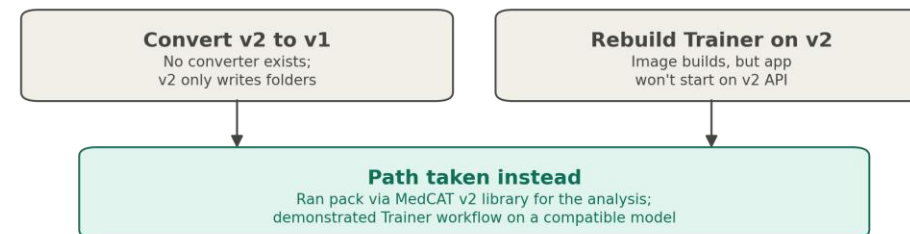
Synonyms



```

graph TD
    A[smoking_findings_only pack  
MedCAT v2 format (cdb/ folder)] --> B[Uploaded to MedCATtrainer  
Trainer runs MedCAT v1.16]
    B --> C[Format mismatch  
v1 expects flat cdb.dat, not a folder]
    C --> D[CDB extracts empty  
~4KB stub vs ~255MB real data]
    D --> E[No concepts imported  
Nothing highlights, search fails]
  
```

Attempted fixes — all blocked



```

graph TD
    A[Convert v2 to v1  
No converter exists;  
v2 only writes folders] --> D[Path taken instead  
Ran pack via MedCAT v2 library for the analysis;  
demonstrated Trainer workflow on a compatible model]
    B[Rebuild Trainer on v2  
Image builds, but app  
won't start on v2 API] --> D
  
```

Root cause: Trainer (v1) not yet compatible with v2 model packs

Limitations & Path to Production

Where it currently breaks (limitations)

- **No negation/status model** - the pack has no MetaCAT, so "denies smoking," "no tobacco," and discharge advice ("smoking cessation") get mis-flagged; handled with **hand-written regex rules** as a stopgap - which miss edge cases like "no longer smokes" (→ former, not never)
- **False positives** - substring matches like "ex" in "ex-lap" → Ex-smoker, "every day" → Daily
- **61 conflicting records** - ~12% of admissions had contradictory mentions across notes
- **452 unknown (45%)** - no smoking evidence in the discharge summary; honestly flagged rather than guessed

How to improve (next steps)

- **Add a trained MetaCAT status model** - learns affirmed vs negated vs hypothetical, replacing the rules
- **Tighten concept filtering** - restrict to genuine smoking SNOMED concepts to kill false positives
- **Widen beyond discharge summaries** - process physician/nursing notes to recover unknowns
- **Capture temporality & pack-years** - "quit 20 yrs ago" vs "quit last week" both read as "former" today
- **Validate against clinician annotation** - produce real precision/recall/F1 before production use

Thank you